



Seat No: _____

No. of Printed Pages: 02

116

SARDAR PATEL UNIVERSITY
F.Y.B.C.A – 2nd Semester Examination -2023
US02CBCA51 || Advanced C Programming

Date : 15/04/2023, Saturday
Time :02:00 p.m. to 05:00 p.m.

Total Marks: 70

Q.1 Select the correct option for each of the followings:

10

- 1 In prototype declaration, specifying _____ is optional.
(a) parameter name (b) data type (c) semicolon (d) return type
- 2 A function which calls itself is known as _____.
(a) reverse (b) recursive (c) reserve (d) none
- 3 The parameters used in a function call are called _____ parameters.
(a) arguments (b) formal (c) actual (d) none
- 4 The variables declared in a structure definition are called as its _____.
(a) members (b) objects (c) record (d) none
- 5 _____ operator connects the structure name to its member.
(a) underscore (b) dot (c) ! (d) none
- 6 Which of the following share a similarity in syntax?
(a) Union & Array (b) Structure & Array (c) Structure & Union (d) None of these
- 7 Which of the following is not a C memory allocation function?
(a) malloc() (b) calloc() (c) realloc() (d) alloc()
- 8 Which of the following defines a pointer variable to an integer?
(a) int &ptr; (b) int **ptr; (c) int &&ptr (d) int *ptr;
- 9 The mode _____ is used for opening a file for updating.
(a) r (b) w (c) r+ (d) a
- 10 When fopen() is not able to open a file, it returns _____.
(a) EOF (b) zero (c) one (d) NULL

Q.2 Explain following in brief. (Attempt any Ten)

20

- 1 Write difference between actual and formal parameter.
- 2 Explain recursion in brief.
- 3 Write advantages of user-defined function.
- 4 Which are the ways to access structure members?
- 5 Which elements are required for compile time initialization of a structure variable?
- 6 Differentiate structure and union.
- 7 List benefits of pointer.
- 8 Give different forms to declare a pointer variable. Which one is preferable?
- 9 List out any 4 scale factor along with the data types.
- 10 Differentiate: printf() and fprintf().
- 11 List out different file modes.
- 12 List basic file operations performed on file.

[P.T.O.]

Q.3 A. List all and explain any three categories of user-defined function with syntax example. [10]

OR

Q.3 A. List and explain components of user defined function with syntax and example. [10]

Q.4 A What is structure? Explain declaring structure variable and structure initialization with syntax and example. [06]

B Explain structure within structure with example. [04]

OR

Q.4 A Explain array of structure and array within structure with example. [06]

B Explain union with syntax and example. [04]

Q.5 A Explain pointer arithmetic expressions in detail. [06]

B Explain calloc() and free() function with syntax and example. [04]

OR

Q.5 A Explain pointer to an array with syntax example. [06]

B Explain malloc() and realloc() function with syntax and example. [04]

Q.6 A Explain getc() and putc() function with syntax and example. [05]

B Explain fopen() and fscanf() function with syntax and example. [05]

OR

Q.6 A Explain error handling during I/O operations. [05]

B Explain getw() and putw() with example. [05]

—X—